



MIDI CLOCK BAR COUNTER & VISUALIZER

Quickstart Guide

Always in Time
Firmware 1.0

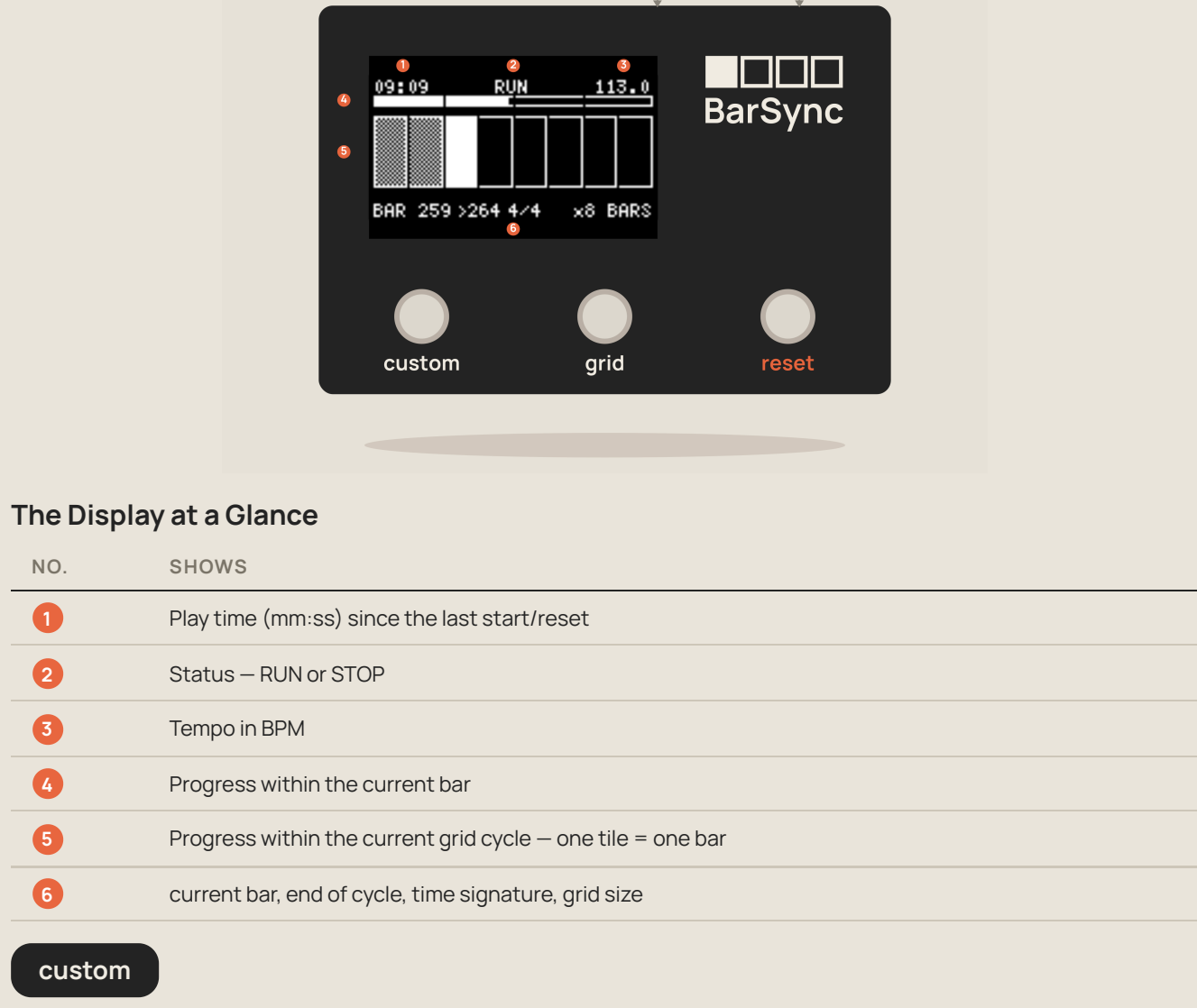
OVERVIEW

What is BarSync?

BarSync keeps you on top of your track's arrangement – live or in the studio. It receives your sequencer's MIDI clock and counts and visualizes the bars elapsed since a starting point you define yourself – either the first MIDI clock signal, or a manual reset via button. That way you always know exactly which bar you're in, and stops, breaks and builds never catch you off guard.

- #### Features
- Bar/beat counter, synced to the incoming MIDI clock (24 PPQN)
 - Grid/progress bar (x1 to x28 bars) – footprint stays constant, tile size adapts automatically
 - Two grid display modes: CYCLE (repeating cycle) or SCROLL (continuous, with markers scrolling into view)
 - Marker Editor: place your own markers on the bar grid (4 symbols) – for arrangement points, phrase starts, or band cues
 - End bar: automatically loop, stop, or continue the count at a defined bar – fixed, or at your last placed marker
 - 5 selectable time signatures (3/4, 4/4, 5/4, 6/8, 7/8)
 - Reset button with two stages (bar end / cycle end), plus a standalone SET 11 function to instantly lay down a new beatgrid (quantized or instant)
 - MIDI Monitor built into the device (Note/CC/PC/Pitch-Bend log, MIDI clock and RUN/STOP graph)
 - Nudge mode to manually compensate for clock phase drift
 - MIDI Thru to pass the signal on to further devices
 - Standby (light sleep) on MIDI inactivity
 - Full settings menu right on the device – no app, no software needed

- ### GETTING STARTED
- #### Connection & First Start
- 1 **Connect your MIDI source.**
Connect your sequencer's DAW interface's MIDI OUT to BarSync's **MIDI IN**.
 - 2 **Optional: pass it on.**
MIDI THRU lets you forward the incoming signal unchanged to another device – handy when BarSync sits in the middle of an existing MIDI chain.
 - 3 **Power it up.**
BarSync starts automatically and shows the retro-style boot screen for about 5 seconds.
 - 4 **Start the MIDI clock.**
As soon as your sequencer is running, the status switches from **STOP** to **RUN**, and the barbeat counter starts running.
 - 5 **Set a reference point (optional).**
The 11 is set automatically as soon as BarSync receives your sequencer's **Start/Run signal** – you don't have to back anything for that. If you'd rather start counting from a specific point instead, just press the **Reset** button.
- If no MIDI clock arrives for a while, the display shows "**NO MIDI CLOCK**" instead – a cue to check your MIDI cable and source (see Troubleshooting on the last page).



- #### The Display at a Glance
- no stop
- 1 Play time (which) since the last start/reset
 - 2 Status – RUN or STOP
 - 3 Tempo in BPM
 - 4 Progress within the current bar
 - 5 Progress within the current gridcycle – one tile = one bar
 - 6 current bar, end of cycle, time signature, grid size
- custom**
- **Short press (SET 11)** – instantly lays down a new beatgrid (default assignment).
 - **Hold 1 second:** toggle MIDI Monitor on/off
 - **Together with Grid:** toggle Nudge mode on/off
 - **Together with Reset (hold 1s):** opens OPERATION SETUP (grid mode, marker editor, end bar)
- grid**
- **Short press:** cycles the number of bars shown in the grid (x1 to x28).
- reset**
- **Short press (Reset 1):** waits for the end of the current bar
 - **Hold (Reset 2):** waits for the end of the current grid cycle
 - **Hold 1 second while booting:** opens the settings menu
 - **On the MIDI Monitor screen:** clears the note/CC list instead
- The Custom button is freely assignable – the functions shown here match the firmware default (Switches - Custom - Function)
- BarSync – Always in Time

- ### OPERATION IN DETAIL
- #### Reset vs. SET 11 – What's the Difference?
- The 11 is set all by itself to begin with: as soon as BarSync receives your sequencer's **Start/Run signal**, counting starts automatically at 11 – without you pressing any button.
- You only need **SET 11** when you want to correct that automatically-set point afterwards – for example when you've started a new song during your live set and your track's 11 no longer "fits" on BarSync's 11. Sure, you could reset the beatgrid and everything would line up again, but during a performance that's obviously something you'd rather avoid. That's where SET 11 comes in: pressing the **Custom** button lays down a new beatgrid with 11 exactly at that moment.
- The two **Reset** stages (1 and 2) build directly on that beatgrid and simply reset the bar counter including the grid (and, depending on settings, the play time) – always in sync with the current beatgrid.
-
- FUNCTION EFFECT**
- | FUNCTION | EFFECT |
|---------------|--|
| Reset | Waits for the end of the current bar (Reset 1) or cycle (Reset 2) and then resets to 11. The beatgrid stays unchanged – the new bar boundaries land exactly where a bar boundary already was. |
| SET 11 | Lays down a completely new beatgrid instantly (Instant) or rounded to the nearest beat (Quantized) – regardless of where the previous beatgrid bar. As a result, every following bar boundary can shift compared to before. |
- Rule of thumb:** If you set runs continuously within the same beatgrid, **Reset** is all you need. Only once a new song section starts with its own downbeat that doesn't match your current count do you need **SET 11** to re-anchor the beatgrid.
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- ### CONFIGURATION
- #### Device Setup
- Open/hold 1s:** hold the Reset button for 1 second while powering on.
- SETUP**
- TIME SIG** – time signature, direct value (3/4, 4/4, 5/4, 6/8, 7/8), default 4/4
- SWITCHES**
- **CUSTOM** – Function (Time sig) / SET 11 – depending on Function, underneath: time-signature selection or Mode (Quantized/Instant)
 - **GRID** – choice of which divisor steps are selectable
 - **RESET** – playtime reset per stage (Real/No)
- DISPLAY**
- **CONTRAST, INVERT** (swap black/white), **GRID LINES** (separators every 32 bars, toggleable)
 - **STANDBY** – on/off, sleep delay
 - **DEFAULTS** – load factory settings (with confirmation)
- Navigating the Menu**
- | BUTTON | FUNCTION IN THE MENU |
|-----------------------------------|-------------------------------|
| Custom | move up one line |
| Grid | move down one line |
| Reset (short) | open selection / change value |
| Reset (hold 1s) | back one level |
| Reset (on page 1, hold 3s) | save, exit menu, restart |

- ### CONFIGURATION
- #### Operation Setup
- Open/hold 1s:** hold **Custom** + **Reset** together for 1 second – during normal operation, not while booting. Holding the same combo again exits the menu from any screen inside it and saves automatically.
- OPERATION SETUP**
- GRID MODE** – CYCLE (repeating cycle) or SCROLL (continuous, markers scroll in from the right)
- SET MARKERS** – opens the marker editor, see below
- END BAR** – OFF / LAST MARKER / MANUAL
- END BAR ACT** – LOOP / STOP / CONTINUE (only shown when END BAR = OFF)
- Navigating the Menu**
- | BUTTON | FUNCTION IN THE MENU |
|------------------------|--|
| Custom | move up one line |
| Grid | move down one line |
| Reset (short) | select the next toggle fit value / GRID MODE, END BAR, END BAR ACT directly, with END BAR = MANUAL + BAR NUMBER (see also above) |
| Reset (hold 1s) | back one level |
- Placing Markers (SET MARKERS)**
- The marker editor lets you tag individual bars with one of four symbols – e.g. for arrangement points, phrase starts, or band cues. It temporarily replaces the beat bar with two navigation arrows.
- | BUTTON | FUNCTION IN THE MARKER EDITOR |
|---------------------------------|---|
| Custom | move the cursor back one bar (left arrow only shown when possible) |
| Grid | move the cursor forward one bar (right arrow only shown when possible) |
| Custom/Grid (hold 1s) | fast page in steps of 16 bars. Custom and Grid are interlocked – only one direction at a time |
| Custom + Grid together | cycles the tool square → circle → triangle → X → DEL (delete) |
| Reset (short) | places the marker with the current tool at the cursor – or erases it when the tool is DEL |
| Custom + Reset (hold 1s) | exits the editor, saves, and returns to OPERATION SETUP |
- Display at the bottom: current bar on the left (with a dividing cycle and number), number of placed markers on the right (out of max. of 500), the current tool, and a "RESET" + "PLACE" + "RESET" + "DELETE" hint in the middle

- ### DIAGNOSTICS
- #### MIDI Monitor
- The built-in MIDI Monitor shows you, right on the device, what's actually arriving on the MIDI line – handy for debugging your setup without any separate software or a computer sitting next to you.
- Show/Hide**
- Hold the Custom button for 1 second (default assignment).
- What's Shown?**
- Event log of the last 6 Note On/Note Off, CC, Program Change and Pitch Bend messages, including channel – newest message on top
 - MIDI clock graph: one spike per detected beat, the downbeat (11) drawn twice as wide as a regular beat
 - MIDI RUN/STOP trace as a level graph, with a plain-text "RUN" + HIGH / "SLUP" + LOW status
- On the Monitor screen, the Reset button takes on a different job:** it clears the note/CC list instead of triggering the normal reset function.
- If no MIDI clock arrives for more than 5 seconds, "NO CLOCK" takes in place of the clock graph instead.**

- ### REFERENCE
- #### Troubleshooting & Quick Reference
- | PROBLEM | SOLUTION |
|---|---|
| Display stays dark | Check power supply, check contrast in the setup menu (Display + Contrast) |
| "NO MIDI CLOCK" is shown | Check the MIDI cable and source, make sure the clock is actually running |
| Buttons don't respond / respond incorrectly | Restart the device; if in doubt, load factory settings (Setup + Defaults) |
| Device only responds after a delay following a long pause | Normal behavior: BarSync goes into standby (light sleep) on MIDI inactivity and wakes automatically on new activity |
- Quick Reference**
- | BUTTON | FUNCTION |
|---------------------------------|--|
| Custom (short) | SET 11 – lay down a new beatgrid* |
| Custom (hold 1s) | toggle MIDI Monitor on/off* |
| Grid (short) | cycle the number of bars shown (x1 to x28) |
| Reset (short/hold) | Reset stage 1/2 |
| Custom + Grid | toggle Nudge mode on/off* |
| Hold Reset while booting | settings menu |
- *The Custom button is freely assignable – this reflects the firmware default.
- Full documentation, schematics, firmware and 3D-printable enclosure:
<https://github.com/benash11/BarSync>